An AI Commons

Social Knowledge, Contributor Compensation, and Shared Governance

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Abstract

Generative artificial intelligence turns accumulated knowledge, expression, and collaboration into services that can be provided repeatedly. This raises a distributional question: what rights should knowledge contributors, developers, and users hold? Using conceptual analysis, normative argument, and institutional comparison, this paper examines the claim that AI should be free because its knowledge comes from humanity. Social origins alone do not establish that all models should be collectively owned or all services permanently free. Such conclusions require additional premises concerning reciprocity, basic capabilities, and accountability. The paper proposes combining compensation for identifiable contributions with public returns, distinguishing licenses for protected works, knowledge in the public domain, model assets, and operational services. It also specifies a budget constraint for basic services. Where firms retain research and operating functions, a public interest asset holder, contributor organizations, and an operating company could use ownership, licenses, and service contracts to secure basic access, benefit sharing, and participation. This is a conditional institutional proposal whose necessity must be assessed against taxation, competition policy, and public procurement. No organizational pilot or effect estimation has been conducted, and no ownership form is presumed inherently superior.

Keywords: generative artificial intelligence; knowledge commons; contributor compensation; public returns; shared governance; basic services

1 Research question and scope

Suppose expert A creates valuable knowledge materials, company C uses relevant materials in developing a model, and user B pays for model services. The intuitive objection is that A supplied the knowledge, yet C is the principal recipient of payment. This connects contributions to production with control in the product market. However, the scenario does not establish whether authorization was required, whether A has already been compensated, what value C added, or whether B is paying for a work license or continuing computation. Without this information, the fairness of a particular transaction cannot be determined.

The object of this paper is generative AI dependent on accumulated social knowledge and the organizations supplying its services. It does not make a uniform ownership claim about every AI technology. The Chinese term gongchan, used in the originating discussion, is examined here as a proposal for common provision and governance: securing durable public interests in access, benefits, and decisions. This limited meaning does not amount to a complete theory of communist society, nor does it presuppose abolishing firms, eliminating all charges, or

creating a single global operator. It retains the political economic question while making its institutional implications assessable.

The paper asks when social contributions justify public returns, how contributor compensation can coexist with broad access, and when these objectives require changes in organizational control rather than taxation, procurement, or regulation of existing firms. Its result is a conditional framework, not an evaluation of an implemented project.

2 From knowledge origins to claims of right

Knowledge expression and model services

The phrase "AI distills human knowledge" can describe the social origins of learning. In technical literature, however, knowledge distillation generally means using the outputs of one model or an ensemble to train another model. This paper calls the broader process the modeling of social knowledge, avoiding the implication that training simply transfers an expert's complete intelligence. Model development also involves algorithms, data preparation, annotation, engineering, and computation. Social knowledge is one input among several. [1]

The content of knowledge must be distinguished from its expression. Ideas, facts, and methods do not automatically become property over which someone can charge indefinitely merely because they can be learned. Particular expressions, contractual obligations, privacy interests, and confidentiality may be protected. WIPO's explanation that copyright protects expression rather than ideas, procedures, and methods supports this distinction. It does not determine whether a particular training activity is lawful, which requires attention to jurisdiction and use. [2], [3]

A's contribution may therefore justify remuneration for labor or a license without establishing a right to collect payment whenever B subsequently uses the model. Conversely, the absence of exclusive rights in particular material does not remove related public interests from institutional debate. Existing legal entitlements and rights that ought to be established are distinct questions.

Social origins and common ownership

Relevant passages in Marx's Grundrisse discuss social knowledge becoming a productive force and its objectification in machinery. They provide a perspective for interpreting how AI turns dispersed knowledge into deployable productive capacities. Applying that perspective to AI is this paper's interpretation. The original text is neither treated as a prediction of large language models nor as a contemporary legal basis for ownership of a specific model. [4]

Moving from social participation in production to a claim that society should share in its returns requires a fairness principle. Moving from public returns to public ownership requires a further argument that less intrusive arrangements, such as taxation, procurement, or licensing, are insufficient. Otherwise, the same reasoning could demand that every activity dependent on language, education, and inherited knowledge be provided without charge. The original claim would then lack a discriminating basis. Social contribution is accepted here as one reason for public claims, not their sole basis or a measure that determines ownership shares.

Shared knowledge and scarce computation

Jones and Tonetti emphasize data's nonrivalry: one party's use does not necessarily reduce the technical possibility of others using the same data. Under their model's assumptions, hoarding can obstruct beneficial sharing. This supports comparing data rights arrangements; it does not imply unconditional disclosure of all data or costless AI inference. [5]

Three levels should therefore be distinguished. Knowledge content can be nonrival. Copying and using models depend on licenses, technical access, and computing resources. Running services occupies equipment, energy, bandwidth, and staff time. A public service can be free at the point of use while still requiring financing. Free access, openness, and common ownership concern price, conditions of reuse, and control respectively; they are not interchangeable.

3 Related research and the paper's contribution

The data as labor approach emphasizes users' roles in producing data and the case for compensation. It helps explain why treating all inputs as resources acquired freely by firms overlooks contributions. Strengthening contributors' bargaining positions still leaves questions about who can establish a contribution, who can negotiate, and how to recognize accumulated knowledge that cannot be allocated to individuals. [6]

Viljoen's relational theory of data governance identifies interests extending beyond individual data subjects that personal rights and separate transactions cannot adequately represent. This paper accordingly treats individual compensation and public governance as distinct. Paying A does not settle whether B, who may have supplied no identifiable data, should have basic access, or how effects on other groups should be addressed. [7]

Benkler's account of commons based peer production provides a perspective on informational collaboration beyond firms and markets. Ostrom's work on polycentric governance indicates that institutions should fit resource characteristics and actual relationships among participants. These works inform institutional comparison here. Findings about irrigation, forests, or open source software are not transferred directly into claims about large model operations. The concentrated computing requirements of foundation model training need separate consideration. [8], [9]

Research on data cooperatives offers resources for member representation and collective bargaining, while data valuation research offers tools for assessing contributions to specified learning tasks. Neither determines just distribution by itself. The authors of Data Shapley explicitly define valuation relative to given data, a learning algorithm, and a performance metric. This paper infers that a contribution score on a particular test set is neither the complete social value of a work nor a final measure of moral entitlement. [10], [11]

The paper does not claim to originate public AI or collective data governance. Its proposed contribution is to examine contributor compensation, basic services, and organizational control together, identifying the financial and accountability conditions for common asset holding and the counterexamples that could weaken its case.

4 Argument and assessment criteria

The paper uses selective, question driven literature searches, prioritizing original papers, author hosted copies, and official institutional explanations. Searches address knowledge nonrivalry, data labor, shared governance, contribution valuation, and public model provision. The evidence register records the relevant sections, abstracts, and source locations

actually inspected through 26 October 2026. This is not a systematic review; no coverage rate is reported, and abstract inspection is not described as verification of an entire paper.

The argument distinguishes descriptive claims from normative premises before asking whether they support an institutional recommendation. Three contestable premises are adopted. Continuing cooperation should give contributors fair opportunities for participation and return. Where a capability affects education, expression, or participation in public life, basic access should not depend entirely on purchasing power. Organizational powers that significantly affect others' opportunities should be subject to appropriate explanation, appeal, and constraints. These are the paper's normative commitments, not measured findings about social consensus.

Institutional comparison proceeds through conditions rather than aggregate rankings. Different mechanisms can pursue the same objective, and comparisons must account for quality, costs, investment, representation, and enforcement. A common holding arrangement that changes only labels, supplies no effective rights, or adds governance costs that substantially impair basic services is not preferable merely because it is described as public.

5 Analytical findings on distribution and governance

Public returns require an additional justification

The first proposition is that the necessary contribution of social knowledge supports deliberation about public returns but cannot determine ownership by itself. Combined with the fairness premises above, it provides a reason to require an account of how public interests are addressed. The strength of that reason also depends on public investment, the importance of a service, users' alternatives, and the effects of organizational power. A limited tool operating under effective competition and a difficult to replace platform performing critical social functions should not automatically face identical obligations.

This addresses the counterexample of a teacher who charges for instruction. Teachers may be paid for additional labor, as may AI developers and operators. Additional public obligations must rest on considerations such as scale, public financing, substitutability, and continuing control, rather than the assertion that machines must not charge because they learn from others. Where these additional conditions do not hold, the case for special ownership arrangements weakens.

Direct compensation and public returns address different claims

The second proposition distinguishes direct compensation from public returns. Direct compensation concerns identifiable and contractible labor, licenses, and other legitimate entitlements. Public returns concern social accumulation that cannot readily be attributed to individuals, as well as broad access and collective participation. They can take the form of basic services, open access to publicly funded outputs, support for underserved languages, or shared governance; they need not all become individual cash dividends.

Under this distinction, a person need not prove inclusion in a training set to qualify for basic access. Equally, serving humanity does not cancel existing payment obligations to particular creators. Contracts should specify the scope of authorization, the form of compensation, and any additional benefit sharing, so that the same payment is not counted twice as a distributional achievement. Material in the public domain should not be registered anew as private knowledge over which an intermediary can collect exclusive fees.

Firms can coexist with public rights

The third proposition is that firms can perform research and operations while public interests are implemented through assets, contracts, and governance rules. Ownership, operating authority, income claims, and participation rights need not belong to one entity. This establishes an institutional possibility, not the availability of the same legal tools in every jurisdiction or its cost advantage over alternatives.

The decisive issue is whether rights affect actual decisions. Advisory seats without information, voting, or remedies do not establish shared governance. Free allowances that an operator can cancel at will do not provide durable basic access. Conversely, where an ordinary commercial firm already meets these requirements through enforceable public service contracts, a separate common holding entity cannot be presumed necessary.

6 Budget conditions for free basic services

To avoid promising free services without identifying who bears their costs, the paper specifies a simplified single period account at fixed service quality. There are N eligible people, each receiving q units of basic service, at incremental resource cost c per unit, with N and c both positive. A unit may be defined as a standardized task; requests of different difficulty cannot simply be treated as identical services. G denotes public support actually available during the period, R commercial revenue, F amortized research, infrastructure, and core staffing costs, C additional operating costs of commercial services, L knowledge and data contributor compensation not already included in F, A governance costs, K agreed financing costs or returns to capital, and V necessary continuity reserves.

Free basic services require, at minimum:

$$G + R \geq F + C + L + A + K + V + Ncq.$$

Holding the other quantities temporarily fixed, the budget permits at most:

$$q \leq \frac{G + R - F - C - L - A - K - V}{Nc}.$$

If the numerator is negative, the organization has a funding gap even with no basic service allowance. If it is zero, the budget cannot support a positive free allowance. Clamping negative values to zero would conceal the underlying deficit and is therefore avoided. Including K does not give capital unconditional priority over labor or basic services; it makes the chosen financing arrangement explicit. Payment priorities must be specified separately in the governance rules.

This relation is only a necessary budget condition under its assumptions, not a commercial forecast or welfare optimum. Revenue may respond to pricing and competition; costs may change with scale, peak demand, and model renewal. Accounting sustainability does not establish adequate quality or capacity. No industry prices, population estimates, or rates of return are substituted into the expression, so it does not estimate a feasible real world allowance.

The constraint nevertheless clarifies a practical choice. Increasing a free allowance requires reliable additional revenue, greater resource efficiency, or transparent changes to other expenditure. Delayed contributor payments, hidden public subsidies, and omitted equipment renewal do not make costs disappear. Basic services must also specify minimum quality,

availability, language coverage, and appeal channels. An ostensibly free entry point that persistently fails to complete tasks does not meet the paper’ s access objective.

7 Shared governance while retaining firms

Asset holding and operating responsibilities

The proposal is best initially considered for new projects, publicly financed outputs, or voluntary contractual arrangements. A public interest entity could lawfully acquire model assets, durable licenses, or shares in an operating company to protect specified public purposes. Representing public interests does not automatically confer rights over historical works or existing private models. Whether the holder is a foundation, cooperative, or another legal entity depends on the jurisdiction; a trust is not assumed to be a universally available solution.

The operating company would undertake engineering, training, deployment, maintenance, and commercial contracting in return for transparent service payments and agreed returns. The asset holder would select operators using public criteria and specify migration, exit, and continuity obligations. Transferable assets must include documentation and permissions needed for actual operation. A model download link without usable deployment conditions does not establish that an operator can be replaced.

Table 1 summarizes the proposed roles. These are institutional design choices made in this paper, not findings from a survey of existing organizations.

Entity	Main responsibility	Required rights or constraints
Public interest asset holder	Hold specified assets and protect public purposes	Disposal constraints, public reporting, accountability
Contributor organization	Negotiate authorization and compensation	Information access, member representation, appeals
Operating company	Develop, deploy, and maintain services	Remuneration, performance audit, orderly handover
User and public representatives	Set basic service requirements and oversee provision	Proposals and votes, quality complaints, replacement of representatives

Representation and decision authority

Shared governance must specify whom representatives represent. Contributor representatives should be selected by the relevant membership. User representation should include both current users and people who remain underserved. Representation for underserved languages and accessibility needs should not depend solely on spending. Terms of office, conflict disclosures, and workable replacement procedures are necessary. Participation also requires compensation for time and intelligible information; otherwise, formally equal opportunities may favor those with the most resources.

A board or equivalent body would decide budgets, basic service standards, major licenses, and asset disposals, leaving routine research to professional teams within these constraints. Decisions ending basic services or transferring important public assets should require separate assent from public and contributor representatives, rather than capital votes alone.

Mediation, reconsideration deadlines, and interim arrangements for maintaining minimum services are also needed to address deadlock. Seat allocations and voting thresholds are variables for subsequent testing, not arbitrary ratios presented as scientific findings.

Governance does not mean deciding scientific facts by majority vote. Model assessments should report reproducible task performance and failures. The governing body chooses objectives and resources; it cannot vote away unfavorable evidence. Auditors need access to related party transactions and internal cost allocations so that rising management fees cannot leave a nominal benefit sharing arrangement permanently without distributable income.

Compensation and the limits of attribution

Different contributions may call for different rules. Commissioned annotation, evaluation, and expert services can be paid under labor or service contracts. Where use of a work legally requires permission, payment is determined by the authorization arrangement; rights holders may also voluntarily license without charge. Recorded retrieval uses can be metered where licenses permit. General capabilities acquired through training, however, cannot be presumed to support accurate attribution of each answer to individual authors. Collective compensation pools may reduce negotiation costs, but membership, calculation rules, and dispute procedures must be transparent.

Valuation is one input into negotiation and verification. Data volume, download counts, or a single benchmark can miss scarce knowledge and long term value. Subsidies for underserved languages or public educational content should state their objectives in advance rather than claim to reflect precise marginal contribution calculations without supporting evidence. Neither individual nor collective authorization displaces data subjects' applicable privacy and other protections.

Exit rules must distinguish stopping future collection, excluding data from later training, and changing models already released. Organizations should not promise deletion effects that their technology and contracts cannot deliver. Common use of knowledge also does not require disclosure of private records, confidential materials, or data whose release could harm particular groups.

Financing and gradual implementation

Basic services may be supported by continuing public procurement, cross subsidies from commercial services, or funds with an explicit duration. A one time donation supports only commitments consistent with its horizon. Budgets should disclose actual basic service costs, contributor payments, related party purchasing, research expenditure, and reserves, allowing users to assess how long current commitments can be maintained.

An initial organization could use a lawful, clearly bounded knowledge collection and models it is entitled to deploy, testing contracts, compensation, and operations before deciding whether foundation model training is necessary. Retrieval services must not be advertised as having solved training data benefit allocation; they provide a more tractable starting point for identifiable contributions. Persistent funding gaps should trigger announced procedures for revising allowances and financing, rather than cuts confined to contributors with weak bargaining power.

8 Alternatives and a practical reference point

Table 2 compares possible mechanisms and conditions requiring examination, without assigning unmeasured scores. There is variation within each category, and names do not establish performance.

Institutional arrangement	Principal mechanism	Conditions requiring scrutiny
Commercial firms under public rules	Taxation, competition constraints, subsidies, or service procurement	Enforcement, switching costs, conditions on public support
Contributor cooperative	Collective licensing and member benefit sharing	Nonmember interests, representation, financing
Direct public provision	Public budgets for research and services	Stable budgets, technical capability, administrative accountability
Public interest holding with commercial operation	Asset constraints and enforceable operating contracts	Governance costs, effective control, continuity arrangements

A hybrid arrangement is more attractive where contracts alone permit lock in, important assets are difficult to replace, and public organizations need firms' operational expertise. Where competition works, switching is easy, and procurement terms can be enforced, purchasing services directly may cost less. Common asset holding is one way to pursue public purposes; support for it here is conditional on these comparisons.

The practical reference is the published record of Apertus. ETH Zurich's official announcement states that EPFL, ETH Zurich, and CSCS released the open language model on 2 September 2025 and describes openness and deployment partnerships. This establishes public institutional participation in model provision. It does not establish lower total costs, compensation for every training contributor, or global user control. This paper has not audited the model's performance, costs, or governance. [12]

Humanity wide common ownership also encounters national boundaries. Knowledge crosses borders, while fiscal responsibilities and enforcement generally belong to particular jurisdictions. A model supported by one country's public budget does not thereby become collectively owned by humanity. Interinstitutional licensing, differentiated financing, and representation across language communities merit examination, but allocating obligations and rights among funding countries, source communities, and external users remains a separate problem.

9 Evaluating the institutional proposal

Subsequent research could begin with a bounded organizational pilot. The following is an unimplemented evaluation design, not results already obtained. The pilot should establish a service domain, intended population, permitted data, and equivalent quality requirements before comparing institutions. If model capabilities, staffing, and subsidies all change together, differences in outcomes cannot be attributed entirely to ownership.

Service assessment should combine total cost per comparable task successfully completed with failure rates, waiting times, and language disparities, rather than count free calls alone. Compensation assessment should report payment coverage, delays, disputes, and concentration. Governance assessment should record who introduces issues, which proposals change decisions, and whether appeal outcomes are enforced. Sustainability assessment should track research investment, reserves, and subsidy dependence. Indicators should be reported separately rather than compressed into an overall fairness score with unjustified weights.

Researchers should specify observations that would weaken the proposal in advance. Examples include persistent additional cost under the same quality and budget constraints without gains in access or contributor position, public representatives unable to affect important decisions, or improved member returns achieved by excluding nonmembers. Such findings should lead to revision or abandonment of the particular design. Where organizational selection and differences in project capability cannot be controlled, comparisons should be presented as descriptive rather than causal. Sample size, duration, and decision thresholds require a separate pilot protocol; none is fabricated here.

10 Limitations

The normative premises are contestable. Different conceptions of justice may support contributor compensation while rejecting universal basic services, or support redistribution through taxes while rejecting shared control. The paper sets out its reasons without claiming that everyone must accept them.

Selective literature retrieval does not establish comprehensive coverage or originality. Complete training records, developer accounts, and licensing contracts were unavailable to this analysis, and no social contribution shares were measured. The paper therefore cannot identify fair equity percentages, a uniform compensation rate, or an optimal free allowance.

The budget expression omits demand responses, intertemporal investment, supply shocks, and policy change; it identifies necessary constraints only. The institutional proposal has not undergone jurisdictional adaptation, actual contract review, or a pilot. Asset restrictions, voting arrangements, and exit rules may require different legal tools in different settings. The Apertus reference supports limited facts about public provision, not the effectiveness of the full proposal.

Training data returns also address only part of AI’s distributional implications. Employment change, environmental costs, supply chain profits, and international capability gaps are not modeled here. Reducing all these issues to payments to data suppliers would omit important interests. Claiming that a basic service subsidy has eliminated inequality would likewise exceed the paper’s conclusions.

11 Conclusion

AI’s dependence on social knowledge raises a continuing institutional question: when common accumulation becomes a saleable capability, how should contributors and the public obtain returns and participation? Origins alone do not establish permanent free provision. Combined with reciprocity, basic capability, and accountability principles, they support conditional claims to public returns. Direct compensation and public returns should be designed together. Appeals to humanity should not displace creators’ legitimate interests,

and inability to demonstrate an individual contribution should not exclude a person from basic services.

The paper proposes a testable arrangement combining public interest asset holding, contributor representation, and commercial operation. It permits remuneration for development and operations while embedding basic services, asset purposes, and important decisions in enforceable rights. Adoption depends on its performance relative to taxation, procurement, cooperatives, or direct public provision. The value of investigating an AI commons lies in translating shared benefit into institutions with identified financing, enforceable obligations, and means of correction.

Materials licensing and reproduction

The complete Chinese and English texts share one paper identifier and content version. The shared bibliography, evidence register, glossary, and revision records are maintained in the local working project; verification records reflect the material actually read. There is no empirical dataset or numerical estimate. Section 6 states the budget variables, derivation, and conditions in full; no subsequent pilot has been conducted. The public release consists of this paper’ s introduction and the complete Chinese and English PDFs.

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AI use statement

The user supplied the research question and initial distributional intuition in discussion. On 26 October 2026, Codex assisted with retrieval, argument organization, drafting in Chinese and English, reference preparation, and formatting checks; the exact model subversion was not independently verified. Checks included accessing relevant original source content, distinguishing normative inferences from empirical claims, comparing the language versions, and inspecting document outputs. These checks were carried out by the same AI assistant and do not constitute independent human verification or external peer review. The author confirmed the byline and release scope for this edition.

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